

Explainable AI Framework for Spam SMS Detection Using Transformers

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Problem Statement

Spam SMS messages are increasingly used for fraud, phishing, malicious links, unwanted advertisements, and social-engineering attacks. Traditional machine learning techniques such as Naïve Bayes, Logistic Regression, and Support Vector Machines (SVM) are commonly used for spam detection, but they mainly depend on word frequencies and extracted features. These methods have limited ability to understand the context and relationships between words in a message.

Furthermore, many existing spam detection systems provide only a Spam or Ham prediction without explaining the reason behind the decision. This lack of transparency can reduce user trust in AI-based security systems.

Therefore, this project proposes an Explainable AI Framework for Spam SMS Detection Using Transformers, which uses BERT for contextual SMS classification and SHAP and LIME to explain the important words contributing to each prediction.

Objectives

- To develop a BERT-based Transformer model for SMS spam detection.
- To classify SMS messages into Spam and Ham categories.
- To improve contextual understanding of SMS messages using BERT.
- To apply SHAP to identify the contribution of important words/features.
- To apply LIME to provide a local explanation for individual predictions.
- To provide a confidence score along with the prediction.
- To generate human-readable explanations for the model's decisions.
- To evaluate the proposed model using Accuracy, Precision, Recall, and F1-Score.

Proposed Framework

SMS Input → Text Preprocessing → BERT Tokenizer → BERT Transformer → Spam/Ham Prediction → SHAP + LIME → Explanation → Prediction

Literature Survey

Existing Method	Technique Used	Advantages	Limitations
Traditional ML	Naïve Bayes	Simple and fast for text classification	Limited contextual understanding
Traditional ML	Logistic Regression	Effective for binary classification	Depends on feature engineering
Traditional ML	SVM	Good performance for text classification	Limited interpretation of predictions
Deep Learning	CNN	Learns important text features automatically	Limited contextual understanding
Deep Learning	LSTM	Captures sequential information	Higher computational requirements
Transformer	BERT	Understands contextual relationships between words	Predictions are difficult to interpret without XAI
Proposed Method	BERT + SHAP + LIME	Context-aware classification with explainable predictions	Requires model training and XAI integration

Dataset:

<https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset>

Research Gap Identification

The literature survey identifies the following research gaps:

1. Traditional machine learning models such as Naïve Bayes, SVM, and TF-IDF-based approaches have limited contextual understanding.
2. CNN and LSTM-based approaches improve automated feature learning but may not capture contextual relationships as effectively as Transformer-based models.
3. Many existing systems focus mainly on classification accuracy.

4. Most systems provide only a Spam/Ham label without clearly explaining the reason behind the prediction.
5. Users need to know which words or parts of a message influenced the model's decision.
6. Existing Transformer-based approaches can provide strong contextual understanding, but there is scope for integrating BERT with multiple explainability techniques in a simple and user-friendly framework.
7. Therefore, the proposed system addresses these limitations by integrating BERT + SHAP + LIME, along with confidence scores and human-readable explanations.

Innovation

- Integration of BERT and Explainable AI for SMS spam detection.
- Use of both SHAP and LIME for explaining predictions.
- Identification of important words influencing Spam/Ham classification.
- Display of prediction confidence along with the classification result.
- Conversion of complex AI decisions into human-readable explanations.
- Provides both detection and explanation in a single framework.

Creativity

- A user can enter an SMS message into the system.
- The system predicts whether the message is Spam or Ham.
- The system provides a confidence score for the prediction.
- Important words influencing the prediction are highlighted.
- SHAP and LIME provide explanations that help the user understand why the message received that classification.
- The system can be presented through a simple web-based interface.

Novelty

The proposed system combines BERT, SHAP, and LIME into a unified SMS spam detection and explanation framework.

Traditional systems generally provide only:

Spam / Ham

The proposed framework aims to provide:

Spam / Ham + Confidence Score + Important Words + Explanation

The novelty of the project is therefore the integration of contextual BERT-based classification with dual Explainable AI techniques, SHAP and LIME, to make SMS spam detection more transparent and understandable.

Feasibility

1. Technical

- The project is technically feasible using modern AI and web technologies.
- Technologies used:
 - Python
 - Hugging Face Transformers
 - BERT
 - SHAP
 - LIME
 - Flask
 - HTML
 - CSS
 - JavaScript
- These open-source tools and frameworks have strong community support and documentation.

2. Economic

- The project is economically feasible and cost-effective.
- Uses open-source tools and libraries.
- Uses freely available datasets from Kaggle.
- No expensive software or licenses are required.
- Development and maintenance costs are low.
- Overall, the project can be developed and deployed with minimal cost.

3. Operational

- The project is operationally feasible and easy to use.
- Provides a simple and intuitive web interface.
- Users can easily enter SMS messages for analysis.
- Provides fast prediction and explanations in real time.
- Can be deployed on a standard computer or cloud server.
- Easy to operate, maintain, and scale for real-world usage.

Overall Feasibility

Technically achievable, economically cost-effective, and operationally practical, making the proposed system highly feasible.

Complete System Workflow

